

Internal Technical Memorandum
Ontology-First Derivation of Mercury Perihelion Precession
Based on the Strüver-Model (Funktionslogik)

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1 Purpose of This Document

This memorandum derives the perihelion precession of Mercury strictly from the ontological foundations of the Strüver-Model, without invoking:

- spacetime curvature,
- metric postulates,
- geometric reinterpretations of gravity,
- event-specific parameter fitting.

The derivation uses only:

1. Energy as transition potential,
2. Mass as energy density,
3. Space as relation between energy densities,
4. Finite stabilization rate of the Φ -operator.

2 Ontological Foundations

2.1 Energy

Energy is defined as transition possibility:

$$E = \text{capacity of state transition}$$

This definition is ontologically prior to geometry.

2.2 Mass

Mass is bound energy density:

$$\rho_E = \frac{dE}{dV}$$

Mass therefore represents a localized concentration of transition potential.

2.3 Space

Space is defined as relational structure between energy densities:

$$\mathcal{R}(r_i, r_j) = f(\rho_E(r_i), \rho_E(r_j))$$

There is no independent geometric substrate. Relations emerge from density contrasts.

2.4 Φ -Operator

Time is not a dimension. Time is the stabilization activity of the Φ -operator.

The Φ -operator restores triadic balance under the 70/30 dynamic stability corridor.
Crucially:

Stabilization is finite-speed.

Define:

$$c_\Phi := \text{maximal stabilization propagation rate}$$

This is not a geometric constant. It is a regulator-bound propagation limit.

3 Classical Central Relation (Newton Limit)

For a central energy density $\rho_E(r)$ generated by the Sun, the leading-order relational response yields:

$$F_N(r) = -\frac{GMm}{r^2}$$

This is the gradient of energy-density relation under instantaneous stabilization.

4 Finite Stabilization Correction

Because Φ stabilizes sequentially (not instantaneously), relation updates propagate with limit c_Φ .

This induces a second-order correction term in the relational field.

Dimensional consistency requires:

$$[\delta a] = \frac{L}{T^2}$$

Only one second-order combination exists using G, M, r, c_Φ :

$$\delta a(r) = -\frac{K}{r^3}$$

with

$$K = \frac{G^2 M^2}{c_\Phi^2}$$

The full acceleration becomes:

$$a(r) = -\frac{GM}{r^2} - \frac{G^2 M^2}{c_\Phi^2 r^3}$$

This term is not fitted. It follows from:

- finite propagation,
- energy-density relational structure,
- dimensional closure.

5 Orbit Equation

Using the Binet substitution:

$$u(\varphi) = \frac{1}{r}$$

The orbital equation becomes:

$$u'' + u = \frac{GM}{h^2} + \frac{3GM}{c_\Phi^2} u^2$$

This nonlinear term prevents closed elliptical orbits.

6 Perihelion Precession

For small correction:

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c_\Phi^2}$$

per revolution.

No event-specific parameter appears.

7 Numerical Evaluation for Mercury

Insert:

$$GM_\odot = 1.32712440018 \times 10^{20} \text{ m}^3/\text{s}^2$$

$$a = 5.79 \times 10^{10} \text{ m}$$

$$e = 0.2056$$

Assume:

$$c_\Phi = c = 2.99792458 \times 10^8 \text{ m/s}$$

Then:

$$\Delta\varpi \approx 43'' \text{ per century}$$

8 Rejection Protocol

1. Fix c_Φ once (a priori).
2. Compute $\Delta\varpi$.
3. Compare to observation.
4. If outside measurement tolerance: reject.

No per-planet fitting allowed.

9 Consistency with Strüver-Model

This derivation:

- preserves triadic logic (Generator: density, Regulator: Φ , Mediator: relational propagation),
- respects 70/30 stabilization corridor,
- maintains ontological closure,
- introduces no geometric substrate,
- requires no spacetime curvature.

10 Conclusion

Perihelion precession emerges as:

Finite-rate stabilization effect

Gravity is not geometry. It is regulated energy-density relation.

If the predicted value matches observation under fixed c_Φ , the model passes first-order solar-system validation.

If not, the model is falsified.

End of Internal Memorandum